Seungsu **Lee**

System Software Engineer

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Education

Seoul National University(SNU) — *Seoul, Korea*

M.S. in Computer Science and Engineering — *Mar. 2019 - Feb. 2021*

- Thesis: Real-time System Optimization of DAG Task Model Considering Applicability
- Real-Time Ubiquitous Systems Laboratory
- Advisor: Prof. Chang-Gun Lee
- GPA: 3.94/4.3

Korea University — *Seoul, Korea*

B.E. in Electrical Engineering — *Mar. 2013 - Feb. 2019*

- Aug. 2015 - Aug. 2017, Mandatory Military Service.
- GPA: 3.98/4.5

Hana Academy Seoul(Hana High school) — *Seoul, Korea*

Mar. 2010 - Feb. 2013

Research Interest

Real-Time Systems / System Software Platforms / Autonomous Driving / Workload Profiling & Optimization

Work & Research

Naver Labs — *Bundang-gu, Seongnam-si, Gyeonggi-do, Korea*

Software Engineer — *Apr. 2022 - Now*

- **Street View Mapping Vehicle Platform**
  Migrated the mapping data-collection vehicle from stock Ubuntu to a custom Yocto-based OS image with read-only rootfs and A/B redundancy, refactoring the system layers and codebase; redesigned the vehicle-control orchestrator as an explicit state machine, reducing startup time by about 44%, and brought the platform to service-level quality through on-vehicle validation and security hardening.
- **Autonomous Shuttle System Software**
  Developed vehicle control, sensor, and UI device drivers; designed and operated the in-vehicle network; integrated new hardware modules into the running system.
- **Release Engineering & Operations**
  Formalized the team's release process (versioning and branching strategy) and operated regular releases; analyzed field failures down to root cause across storage, network, and application layers.
- **Multi-board Yocto Platform for Robotics**
  Built Yocto OS images for various boards (e.g., NVIDIA Orin) used in robot projects; ported a vendor-provided Ubuntu camera driver into the Yocto environment at the kernel level.
- **Real-Time Linux Research**
  Studying PREEMPT_RT-based real-time Linux and scheduling with observability tools (rtla, timerlat, cyclictest) for next-generation platforms.

Samsung Electronics, Device Solutions Division
Hwaseong-si, Gyeonggi-do, Korea

Engineer
Feb. 2021 - Mar. 2022

- **PreemptRT Patch Support**
  Understanding the PreemptRT Patch characteristics and measuring the achievable scheduling latency gain.
- **PCIe Device Driver**
  Developing and bring up Linux Kernel Driver.
- **Enhancement of Code Coverage**
  Configure VectorCAST-based environment and improve code coverage.
- **Debugging with Trace32**
  Debugging in RAMDUMP and target image using Trace32 and iTSP.
- **Building Test Automation Environment**
  Configuration of test automation environment.

Real-Time Ubiquitous Systems Laboratory, SNU
Seoul, Korea

Graduate Researcher(M.S.)
Jan. 2019 - Feb. 2021

- Advisor: Prof. Chang-Gun Lee
- **Autonomous Driving System Optimization Research on Embedded Boards**
  To minimize hardware computing unit cost and software development cost, we build automotive system using embedded board and open source. After analyzing the resource usage behavior measured by the profiling tool, an optimization algorithm is devised to ensure the real-time execution of the task. (Summarized Picture Linked)
- **ROS Profiling Tool Research**
  Understanding the structure and characteristics of ROS and finding the reason why resource usage was not accurately known in the existing profiling tool. Through this, the ROS-based resource use profiling tool is developed. (Summarized Picture Linked)
- **Building an Autonomous Driving Experiment Platform With Embedded boards**
  For scheduling and optimization research, an autonomous driving experiment platform was built in two places: an RC car(1/10 the size of a real car) and a real car. (Summarized Picture Linked)
- **Optimal Task Parallelization Research**
  Targeting Global-EDF scheduling, research on a conditional optimal algorithm for parallelizing tasks with parallelization freedom. (Summarized Picture Linked)
- **ROS Lectures for New Members**
  Create ROS lectures for new members on Youtube(linked).

Advanced Control Systems Laboratory, Korea University
Seoul, Korea

Undergraduate Researcher
Nov. 2017 - Sep. 2018

- **Autonomous Driving Study**
  Studying autonomous driving topics regularly in undergraduate study groups.
  Topics: autonomous driving level, sensors(GPS, LiDAR), linear algebra, image preprocessing

Publication

International Conference

2019	Youngeun Cho, Do Hyung Kim, Daechul Park, **Seungsu Lee**, and Chang-Gun Lee, Conditionally Optimal Task Parallelization for Global EDF on Multi-core Systems, in IEEE Real-Time Systems Symposium	*RTSS 2019*

International Journal

2022	Youngeun Cho, Do Hyung Kim, Daechul Park, **Seungsu Lee**, and Chang-Gun Lee, Optimal Parallelization of Single/Multi-Segment Real-Time Tasks for Global EDF, in IEEE Transactions on Computers	*TC 2022*

Domestic Conference

2020	Jongwoo Han, **Seungsu Lee**, Seonghyeon Park, and Chang-Gun Lee, HSFS : Heterogeneous Superiority-First Scheduling for DAG Task, in Korea Software Congress	*KSC 2020*
2020	Jongwoo Han, Sehwan Lee, **Seungsu Lee**, Youngeun Cho, and Chang-Gun Lee, A Precise Module Execution Time Measurement of ROS Network, in Korea Computer Congress	*KCC 2020*
2019	**Seungsu Lee**, Daechul Park, Youngeun Cho, and Chang-Gun Lee, Necessary Bound of Resource Utilization for Fail-Safe Execution of ROS Task, in Korea Computer Congress	*KCC 2019*

Honors & Awards

SNU-SEC Semiconductor Special Education Program(SSSP) *Samsung Electronics*

FULL SCHOLARSHIP *Mar. 2019 - Feb. 2021*

- Funded full-tuition scholarship with stipend for graduate studies by Samsung Electronics(SEC)

Academic Excellence Scholarship *SNU*

PARTIAL SCHOLARSHIP *Mar. 2019 - Aug. 2019*

- Funded one semester tuition scholarship for graduate studies by Seoul National University(SNU)

The National Scholarship for Science and Engineering *KOSAF*

FULL SCHOLARSHIP *Mar. 2013 - Feb. 2019*

- Funded full-tuition scholarship for undergraduate studies by Korea Student Aid Foundation(KOSAF)

President's List *Korea University*

OUTSTANDING SEMESTER GRADES *Spring & Fall 2013, Spring 2018*

- Received a citation for academic excellence in semester

The Commissioner's Citation *Seoul RMMAO*

EXEMPLARY SERVICE OF SOCIAL WORK PERSONNEL *Aug. 2017*

- Selected as an Exemplary Service of Social Work Personnel by performing military service with a strong sense of mission and responsibility at Seoul Regional Military Manpower Administration Office(Seoul RMMAO)

Experience

TEACHING

2020	Teaching Assistant, M1522.000700: Logic Design	*SNU*
2019	Teaching Assistant, M1522.000700: Logic Design	*SNU*
2019	Teaching Assistant, 400.320: Engineering Research Practice 1 (Undergraduate Research Opportunities Program)	*SNU*

ATTENDING CONFERENCE/WORKSHOP

Dec. 2019	Participants, Korea Software Congress	*Pyeongchang, Republic of Korea*
Aug. 2019	Teaching Assistant, Autoware Workshop	*Chung-ang Univ., Republic of Korea*
Jul. 2019	Participants, Conference on Ph.D Research in Microelectronics and Electronics	*Lausanne, Switzerland*
Jul. 2019	Participants, International Conference on Transportation Information and Safety	*Liverpool, UK*
Jun. 2019	Poster Presenter, Korea Computer Congress	*Jeju, Republic of Korea*

Extracurricular Activity

The KorMent Program *KOSAF*

MENTEE *Mar. 2015 - Nov. 2015*

- Participated in mentoring to learn wisdom, leadership and spirit of service from the mentor
- Mentor: Hanho Kim(Former Vice President of Hewlett-Packard Korea)

Skills

Languages	Korean(Native), English(TOEIC: 905, OPIc: IH)
Programming	C/C++, Python, Shell Script
Systems	Linux, Yocto, Docker, Robot Operating System(ROS)/Autoware, PREEMPT_RT, A/B OTA
Profiling & Tools	perf, ftrace, kernelshark, Wireshark, rtla/cyclictest, Git, Jenkins, Gerrit, LaTeX